BS (Artificial Intelligence) – F23

Fall-2026

Course Title:	Generative Artificial Intelligence
Course Code:	ARI-406
Credit Hours:	(3+0)
Contact Hours:	3 lectures per week
Course Type:	AI Domain Core
Course Instructor:	Rajesh Kumar, PhD
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Description:

Generative Artificial Intelligence introduces students to the principles, architectures, techniques, and applications of modern AI systems capable of generating text, images, code, audio, and other forms of content. The course develops conceptual understanding of generative modelling while placing particular emphasis on foundation models and Large Language Models (LLMs). Students study neural generative models, Transformers, tokenization, language modelling, pre-training, instruction tuning, parameter-efficient fine-tuning, prompt engineering, embeddings, Retrieval-Augmented Generation (RAG), tool use, AI agents, diffusion models, and multimodal generative systems. The course also develops the ability to evaluate generative AI systems in terms of output quality, hallucination, reliability, robustness, computational efficiency, bias, privacy, security, and responsible use. Through assignments, demonstrations, case studies, and a semester project, students design and evaluate practical Generative AI applications using contemporary models and development frameworks. As this course carries no dedicated laboratory session (3+0), hands-on practice is embedded directly within lecture time rather than held in a separate lab period.

Aims and Objectives:

- Explain the principles of generative modelling and distinguish generative models from discriminative approaches.
- Explain the architecture and operation of Transformers and Large Language Models.
- Understand tokenization, embeddings, attention, pre-training, inference, and post-training of foundation models.
- Design effective prompts and systematically evaluate prompt-based solutions.
- Apply pre-trained generative models to practical AI problems.
- Develop applications using embeddings, vector retrieval and Retrieval-Augmented Generation.
- Understand fine-tuning, instruction tuning and parameter-efficient adaptation methods.
- Design basic tool-using and agentic AI systems.
- Explain diffusion-based and multimodal generative models.
- Evaluate Generative AI systems for accuracy, hallucination, robustness, safety and efficiency.
- Identify ethical, privacy, security, copyright and social issues associated with Generative AI.
- Complete an end-to-end Generative AI project addressing a real-world problem.

Note: These objectives support the three prescribed CLOs but do not replace or modify them.

Prerequisites:

Prerequisite: Machine Learning.

Recommended Background: Python programming, basic Deep Learning, probability and statistics, linear algebra, and introductory Natural Language Processing concepts.

Assessment:

S. No	Assessment Activities	Percentage	Sub-total	Activities
1a	Sessional: Quizzes, Assignments & Case Studies	20%		6
1b	Semester Project (end-to-end GenAI application)	10%	30%	1
2	Midterm Examination	30%	30%	1
3	Final Examination	40%	40%	1
	Total		**100%**	

Sessional Total: 30% (20% Quizzes/Assignments/Case Studies + 10% Semester Project).

Assessment-CLO Emphasis:

Assessment	CLO-1	CLO-2	CLO-3
Quizzes	✓		✓
Written / Programming Assignments	✓	✓	✓
Case Studies	✓	✓	✓
Midterm Examination	✓	✓	✓
Semester Project		✓	✓
Final Examination	✓	✓	✓

Course Learning Outcomes (CLOs):

No.	Course Learning Outcome	Domain	GA	Assessment Tools
CLO-1	Analyze the principles, architectures and learning mechanisms of generative models, foundation models, LLMs and multimodal generative systems.	C4	GA-3	Quizzes, Assignments, Midterm, Final Exam
CLO-2	Develop Generative AI solutions using prompting, pre-trained models, embeddings, RAG, model adaptation, and agentic techniques.	C6	GA-4	Assignments, Case Studies, Semester Project
CLO-3	Evaluate Generative AI systems in terms of performance, reliability, safety, ethical considerations, and responsible use.	C5	GA-9	Assignments, Project, Midterm, Final Exam

C = Cognitive Domain (C1: Remembering, C2: Understanding, C3: Applying, C4: Analyzing, C5: Evaluating, C6: Creating).

Note: GA mapping follows the approved BS Artificial Intelligence program Graduate Attributes / PLO framework.

CLO–Graduate Attribute Mapping:

CLO	Graduate Attribute	Justification
CLO-1	GA-3: Problem Analysis	Understanding and comparing generative architectures requires analysis of model structure, learning objectives, data characteristics and application requirements.
CLO-2	GA-4: Design/Development of Solutions	Developing prompt-based, RAG, adapted-model and agentic solutions requires students to design and evaluate computational solutions for defined problems.
CLO-3	GA-9: Ethics (related: GA-8: Computing Professionalism and Society)	Evaluating bias, privacy, security and responsible use requires the graduate to understand and commit to professional ethics and to assess the societal, legal and cultural impact of computing systems.

Tentative 16-Week Teaching Plan:

Week	Lecture 1 – 50 Minutes	Lecture 2 – 50 Minutes	Lecture 3 – 50 Minutes	Principal CLO
1	**Introduction to Generative AI** AI/ML/DL vs. Generative AI; generative vs. discriminative modelling; what GenAI can generate: text, image, audio, video, code; representative applications (2.08%)	**Evolution of Generative Models** statistical generative models; autoencoders; VAEs; GANs; Transformers; diffusion models; foundation models (4.17%)	**Foundation Models and the GenAI Ecosystem** foundation models; LLMs vs. SLMs; open vs. proprietary models; training/adaptation/inference; GenAI application lifecycle (6.25%)	CLO-1
2	**Probability in Generative Modelling** probability distributions; joint/conditional probability; maximum likelihood; learning a data distribution; sampling (8.33%)	**Autoregressive Generative Models** sequence probability; conditional generation; next-token prediction; autoregressive decoding; sampling strategies (10.42%)	**Classical Neural Generative Models: Autoencoders, VAEs and GANs** principles and latent spaces (conceptual treatment); representation learning; strengths and limitations relative to modern approaches (12.50%)	CLO-1
3	**From Text to Machine Representation** text preprocessing; vocabulary; tokens & token IDs; embeddings (14.58%)	**Tokenization** word-level & character-level tokenization; subword tokenization; Byte Pair Encoding; WordPiece/SentencePiece; vocabulary-size trade-offs (16.67%)	**Language Modelling** language models; context; next-token prediction; probability over sequences; context windows; generation process (18.75%)	CLO-1
4	**Attention** limitations of recurrent models; Query/Key/Value; attention scores; scaled dot-product attention; self-attention (20.83%)	**Transformer Components** multi-head attention; feed-forward networks; residual connections; normalization; positional information (22.92%)	**Transformer Architectures** encoder-only models; decoder-only models; encoder-decoder models; causal attention; Transformer inference (25.00%)	CLO-1, CLO-2

Week	Lecture 1 – 50 Minutes	Lecture 2 – 50 Minutes	Lecture 3 – 50 Minutes	Principal CLO
5	**Large Language Model Architecture** parameters; model depth and width; embedding layers; Transformer blocks; output logits and probabilities (27.08%)	**LLM Pre-training** pre-training data collection & cleaning; pre-training objectives; self-supervised learning; training pipeline (29.17%)	**Scaling and Model Behaviour** model size, data scale, compute; scaling considerations; capabilities and limitations; training vs. inference (31.25%)	CLO-1
6	**Fine-Tuning** pre-training vs. fine-tuning; supervised fine-tuning; domain adaptation; instruction tuning (33.33%)	**Parameter-Efficient Fine-Tuning** motivation for PEFT; adapter-based methods; LoRA; quantized adaptation concepts; benefits and limitations (35.42%)	**Alignment and Preference Optimization** human feedback; RLHF conceptual workflow; reward models & preference datasets; DPO concepts; alignment challenges (37.50%)	CLO-1, CLO-2
7	**Prompt Engineering Fundamentals** prompt structure: instructions, context, examples, constraints, output format; zero-shot and few-shot prompting (39.58%)	**Advanced Prompting** prompt templates; decomposition; self-consistency concepts; structured reasoning prompts; prompt chaining & iterative refinement (41.67%)	**Context Engineering and Prompt Evaluation** context-window management; selecting relevant context; prompt experiments & evaluation criteria; prompt injection introduction; robust prompt design (43.75%)	CLO-2, CLO-3
8	**Embeddings and Semantic Similarity** dense vector representations; sentence/document embeddings; similarity measures; semantic search; embedding applications (45.83%)	**Midterm Examination** (47.92%)	**Midterm Feedback and RAG Introduction** diagnostic feedback; limitations of model-internal knowledge; external knowledge; introduction to Retrieval-Augmented Generation (50.00%)	CLO-1 to CLO-3
9	**RAG Architecture** documents & chunking; embeddings & vector databases; retrieval; prompt augmentation; generation (52.08%)	**Building a RAG Pipeline** document ingestion; chunk size and overlap; embedding generation & indexing; top-k retrieval; context construction (54.17%)	**Advanced RAG and Evaluation** query rewriting; metadata filtering; reranking; hybrid retrieval; retrieval evaluation: faithfulness & groundedness (56.25%)	CLO-2, CLO-3
10	**LLM Application Architecture** model interfaces: APIs vs. open-source; system/user instructions; structured outputs; application pipelines (58.33%)	**Tool and Function Calling** external tools; function schemas; structured arguments; tool selection; connecting models to external systems (60.42%)	**Reliability and Guardrails** input/output validation; schema enforcement; guardrails; fallback mechanisms; human-in-the-loop systems (62.50%)	CLO-2, CLO-3
11	**Introduction to AI Agents** LLM vs. agent; goals, environment, memory; tools; actions and observations (64.58%)	**Reasoning, Planning and Tool Use** task decomposition; planning; tool selection; execution loops; reflection concepts; error recovery (66.67%)	**Agentic Architectures** single-agent systems; multi-agent systems; agent communication; workflow-based agents; limitations & failure modes; agent evaluation (68.75%)	CLO-2, CLO-3
12	**Generative Image Modelling** GAN recap; limitations of GANs; introduction to diffusion; noise and denoising (70.83%)	**Diffusion Models** forward diffusion; reverse diffusion; noise prediction; sampling; conditional generation (72.92%)	**Text-to-Image Generation** text conditioning; latent diffusion; guidance; image generation pipelines; image editing applications (75.00%)	CLO-1, CLO-2
13	**Multimodal Foundation Models** text, image, audio, video modalities; cross-modal representations (77.08%)	**Vision-Language Models** image encoding; text encoding; vision-language interaction; visual question answering; image understanding & generation (79.17%)	**Multimodal Applications** document understanding; image reasoning; speech and audio generation; multimodal assistants; case studies (81.25%)	CLO-1, CLO-2

Week	Lecture 1 – 50 Minutes	Lecture 2 – 50 Minutes	Lecture 3 – 50 Minutes	Principal CLO
14	**Evaluation of Generative AI** task-based evaluation; reference-based metrics; human evaluation; model-based evaluation; benchmarking & its limitations (83.33%)	**Hallucination, Robustness and Security** hallucination & grounding; prompt injection; jailbreaking; adversarial inputs; data leakage; red-team testing (85.42%)	**Responsible Generative AI** bias and fairness; privacy; transparency and accountability; intellectual-property considerations; misuse and misinformation; responsible deployment (87.50%)	CLO-3
15	**Efficient Generative AI** model size vs. performance; distillation; quantization; efficient inference; Small Language Models (89.58%)	**Deployment Considerations** local vs. cloud deployment; latency, throughput, memory, cost; model selection; monitoring (91.67%)	**End-to-End GenAI Case Study** problem definition; model selection; prompt/RAG/agent design; evaluation; safety analysis; deployment considerations (93.75%)	CLO-2, CLO-3
16	**Student Project Presentations** problem definition; proposed architecture; implementation; demonstration; results (95.83%)	**Project Evaluation and Peer Review** technical evaluation; GenAI evaluation criteria; limitations; responsible-use considerations; peer feedback (97.92%)	**Emerging Directions and Course Review** emerging foundation-model architectures; efficient/specialized models; long-context & multimodal AI; agentic AI research directions; CLO review & final-exam preparation (100.00%)	CLO-1 to CLO-3

Suggested Applied Assignments:

Assignment	Focus	Suggested Week
1. Tokenization and Language Modelling	Investigate tokenization, context windows and next-token prediction using a small language model.	Week 3
2. Prompt Engineering	Design and systematically compare prompts for summarization, classification, information extraction or question answering.	Week 7
3. Retrieval-Augmented Generation	Develop a small document-question-answering system using embeddings, vector retrieval and an LLM.	Week 9
4. Model Adaptation or Agentic AI	Experiment with parameter-efficient model adaptation, or develop a basic tool-using AI agent.	Weeks 6, 10–11

Semester Project:

Students design an end-to-end Generative AI application addressing a real-world problem. The project carries an explicit 10% weight (see Assessment) in recognition of its scope. Possible projects include:

- University information assistant using RAG
- Research-paper question-answering system
- Generative AI educational tutor
- Multilingual information assistant
- AI document-analysis system
- Code-generation assistant
- Domain-specific chatbot
- AI agent using external tools
- Multimodal document assistant
- Responsible-AI evaluation system

The project report and demonstration should include:

1. Problem definition
2. Dataset / knowledge-source description
3. Model selection
4. System architecture
5. Prompt / RAG / agent design
6. Implementation
7. Evaluation methodology
8. Results
9. Error and hallucination analysis
10. Safety and responsible-use analysis
11. Limitations
12. Final demonstration and report

Recommended Software and Development Resources:

Resource	Use
Python / Jupyter / Google Colab	Core language and free-compute notebooks for all in-lecture hands-on work (no dedicated lab slot).
PyTorch	Underlying deep-learning framework for model internals (Weeks 4–6, 12).
Hugging Face Transformers, Datasets & Tokenizers	Pretrained models, datasets and tokenization pipelines (Weeks 3–8).
Sentence Transformers	Embeddings for semantic search and RAG (Weeks 8–9).
Hugging Face Diffusers	Diffusion / text-to-image pipelines (Week 12).
FAISS (or equivalent vector-search system)	Vector indexing and retrieval for RAG (Week 9).
Git and GitHub	Version control for assignments and the semester project.
Open-source LLMs and/or approved LLM APIs	Inference, fine-tuning and application-building throughout the course.

The course should remain framework-independent wherever possible so that concepts remain useful as specific Generative AI tools change.

Recommended Textbooks:

- **Raschka, S. Build a Large Language Model (From Scratch). Manning.**
- **Foster, D. Generative Deep Learning (2nd ed.). O'Reilly.**
- **Tunstall, L., von Werra, L., & Wolf, T. Natural Language Processing with Transformers. O'Reilly.**
- **Jurafsky, D., & Martin, J. H. Speech and Language Processing. Selected chapters on language models, Transformers and Large Language Models.**

Recommended Free Online Learning Resources:

Resource	Recommended Use
Stanford CS336 – Language Modeling from Scratch	Deeper, implementation-focused treatment of tokenization, pre-training, Transformers, scaling and evaluation — useful for the instructor and Weeks 3–5, 14.
Hugging Face LLM Course	Transformer models, tokenizers, pretrained models, fine-tuning and inference — matches Weeks 3–6, 8.
Hugging Face Agents Course	AI agents, tools, reasoning and planning — matches Weeks 10–11.
Hugging Face Diffusion Models Course	Diffusion theory, image generation and conditional generation — matches Week 12.
Google Machine Learning – Introduction to Large Language Models	Tokens, Transformers, self-attention, LLM training, fine-tuning and distillation — supplementary for Weeks 3–6, 15.
Berkeley LLM / Agentic AI course material	LLM foundations, RAG, reasoning, planning, tool use, agents, multimodal agents, evaluation, safety and ethics — supplementary throughout.

Recommended Research-Paper Reading Categories:

Students should be introduced to selected foundational papers covering:

- Transformer architecture
- Large-scale language models
- Instruction tuning
- Human-feedback alignment
- Parameter-efficient fine-tuning
- Retrieval-Augmented Generation
- Diffusion models
- Multimodal foundation models
- Generative AI evaluation
- LLM agents
- AI safety and responsible Generative AI

Course Rationale

Generative Artificial Intelligence has become a foundational capability across the computing industry, and BS Artificial Intelligence graduates are expected to understand both the underlying mechanisms of generative systems and how to apply them responsibly. ARI-406 accordingly progresses from generative-modelling fundamentals through Transformer-based foundation models to applied techniques (prompting, retrieval-augmented generation, model adaptation and agentic systems), concluding with evaluation, safety and responsible-deployment considerations, so that graduates can both build and critically assess contemporary Generative AI systems.

This progression is summarised below:

Generative Modelling → Transformers → Foundation Models → LLMs → Prompting → Model Adaptation → Embeddings → RAG → Tool Use → AI Agents → Diffusion → Multimodal GenAI → Evaluation → Responsible Deployment.